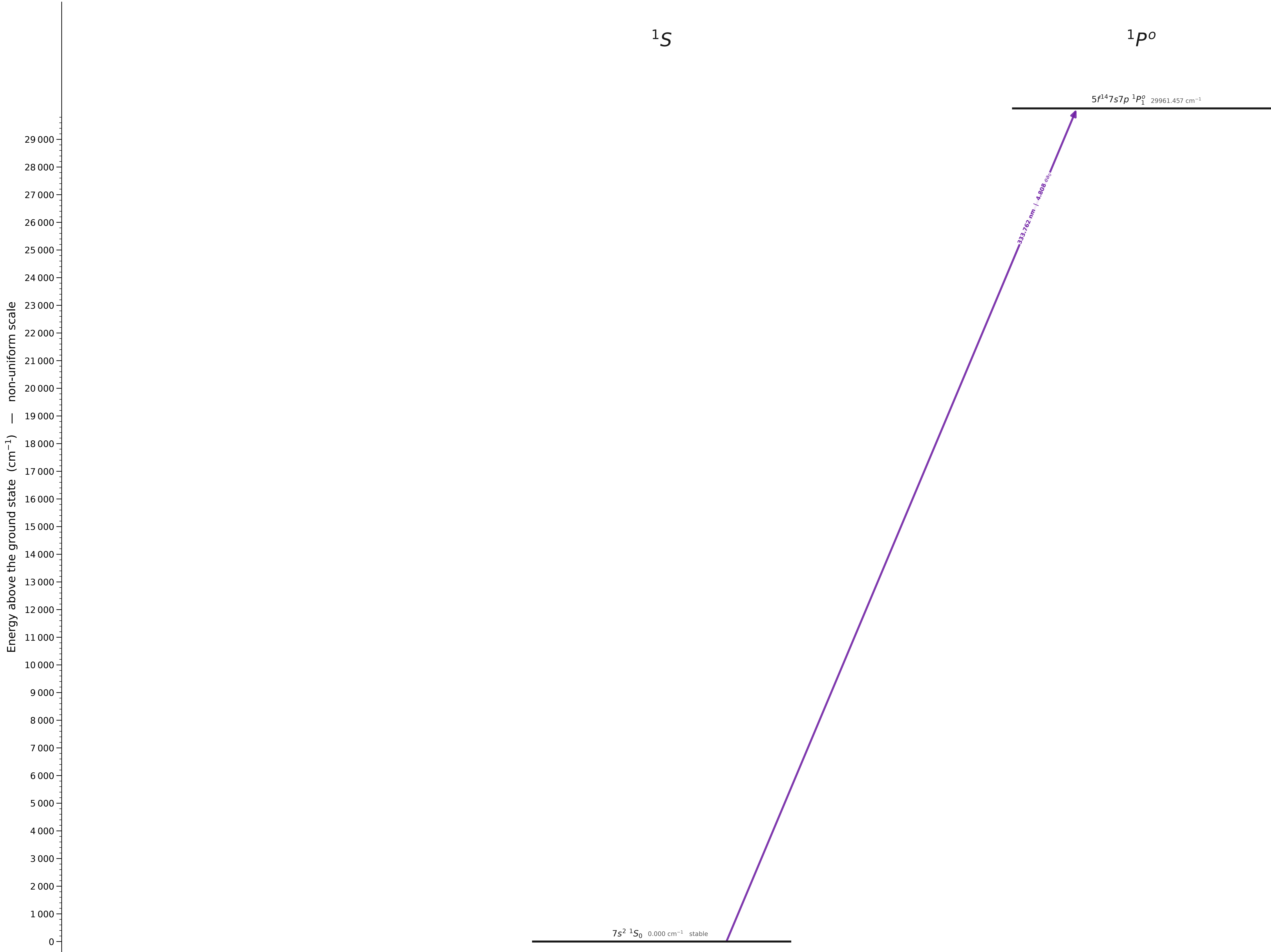


Nobelium

No (No I). Classified transitions below 2 μm (levels and lines from the literature): 2 levels, 1 lines.

No⁺ ionisation limit 53444.00 cm⁻¹ = 6.62621 eV (187.112 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$7s^2\ ^1S_0 - 5f^{14}7s7p\ ^1P_1^o$	333.7621	333.6662	898.2219(12)	4.808	4.200e+08	-	-	the only optical line of nobelium observed so far: first...

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$7s^2\ ^1S_0 - 5f^{14}7s7p\ ^1P_1^o$	333.762	253-254	6720(180)
$7s^2\ ^1S_0 - 5f^{14}7s7p\ ^1P_1^o$	333.762	252-254	10080(690)
$7s^2\ ^1S_0 - 5f^{14}7s7p\ ^1P_1^o$	333.762	255-254	-7700(800)

Shift = ν(first isotope) − ν(second isotope).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

²⁵³No hyperfine structure (*I* = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$5f^{14}7s7p\ ^1P_1^o$	734(46)	2820(690)	F=7/2: -2744.50 F=9/2: -2614.00 F=11/2: +4008.00

Measured A, B constants; sources in the data file.